

AirFare

Real-time Airfare Price Index for India

Product Requirements Document · Version 2.0

A statistically rigorous, auditable price measurement system that replaces manual airfare surveying with automated high-frequency collection, and measures what Indian travellers actually pay rather than what airlines first advertise.

Field	Detail
Product name	APIx - Real-time Airfare Price Index
Document type	Product Requirements Document (PRD)
Version	2.0 (supersedes the v1 System Specification)
Programme	Smart India Hackathon - MoSPI Problem Statement
Primary users	National Statistical Office (NSO), Reserve Bank of India, DGCA / Ministry of Civil Aviation
Secondary users	Economic researchers, aviation analysts, general public
Status	Approved for prototype build

Contents

1. Executive Summary

India measures retail inflation through the Consumer Price Index. Inside that index sits air travel, and the way its price is collected today has not kept pace with how Indians actually buy flights. Prices are gathered by hand, from a small number of physical ticket counters, on a fixed monthly cycle. More than nine out of ten domestic tickets are now bought online, at prices that change several times a day.

The result is a measurement blind spot. The official number is slow, thin, and increasingly disconnected from the fare a traveller sees on their phone.

APIx closes that gap. It is an automated price measurement platform that collects airfares continuously from airline and aggregator platforms, strips them down to their true components, corrects them using established statistical methods, and publishes a daily airfare inflation index that NSO can feed into the CPI and RBI can read as a live signal.

What makes APIx different in one sentence

Everyone else can scrape a fare. APIx is the only system that converts a scraped advertised price into a statistically valid consumer price, and can prove every published number back to its raw source.

The five outcomes we commit to

- **Daily instead of monthly.** A published airfare index every single day, with zero reporting lag, against the current monthly cycle.
- **Full price instead of headline price.** The index captures convenience fees, baggage, seat charges and taxes, not just the advertised base fare.
- **What people pay, not what is quoted.** A correction layer converts advertised offer prices into estimated realised transaction prices.
- **Explainable movement.** Every change in the index can be broken down into which routes, carriers, fees and booking windows caused it.
- **Fully auditable.** Any published figure can be independently recomputed from stored raw data by a third party.

2. Problem Statement

2.1 The situation today

The Consumer Price Index is the official measure of retail inflation in India. The Reserve Bank of India uses it to set interest rates under the inflation targeting framework. Within the CPI, the Transport and Communication sub-group includes air travel.

Air travel prices for this sub-group are collected the way most CPI items have always been collected: field investigators visit a fixed set of outlets and ticketing offices on a set schedule and record the price they are quoted. That method works well for a bar of soap. It works badly for an airline seat.

2.2 Why airline seats break the traditional method

An airline seat is unlike almost every other item in the CPI basket in four ways.

Characteristic	Why it defeats manual monthly collection
The price moves constantly	Airlines use automated pricing software that changes fares many times a day based on how fast seats are selling. The same seat on the same flight can differ by two to four times between a quiet weekday morning and a festival evening. A single monthly reading cannot represent this.
The price depends on when you book	A ticket bought a day before departure and the same ticket bought 45 days before are, in pricing terms, two different products. Manual collection does not systematically sample across booking horizons.
The advertised price is not the final price	Search results routinely exclude convenience fees, baggage charges, seat selection and other mandatory add-ons that appear only at the payment stage. Recording the search price understates what the traveller actually spends.
The sales channel has moved online	Over ninety per cent of domestic tickets are now sold through airline websites and online travel aggregators. Counting prices at physical counters samples the wrong marketplace.

2.3 The consequence

Taken together, these produce a persistent measurement error in a component that feeds directly into national monetary policy. The official airfare series is slow to react, understates the true out-of-pocket cost of flying, and cannot explain why it moved when it does move.

This is not a criticism of NSO. It is an infrastructure problem. The collection method was designed for an economy of physical shops, and a piece of that economy has moved somewhere the method cannot reach.

2.4 Who is affected

Stakeholder	What they need	What they get today
NSO (MoSPI)	An accurate, defensible, auditable airfare price series that plugs into the existing CPI structure	A slow manual series with limited coverage of the real market
Reserve Bank of India	An early, high-frequency read on transport inflation pressure	A lagging monthly figure that arrives after the pressure has already built
DGCA / Civil Aviation	Visibility into fare surges during disruptions, festivals and capacity shocks	Reactive complaint-driven monitoring with no systematic baseline
Researchers and public	Open, transparent data on what flying actually costs in India	Commercial platforms with closed methods and commercial incentives

3. Competitive Analysis

Several categories of system already collect airfare data. None of them produce a statistically valid consumer price index, because none of them were built to. Understanding exactly why each falls short is what defines our product.

3.1 The landscape

Category	Examples	What they are actually optimised for	Where they fall short for a price index
Flight meta-search	Google Flights, Skyscanner, Kayak	Sending a user to a booking page as fast as possible	Shows the lowest attention-grabbing fare, stops before checkout, so mandatory fees are never captured. No history, no weighting, no methodology.
Consumer price prediction	Hopper, Ixigo price alerts, Google price insights	Persuading a user to book now or buy a price-freeze product	Proprietary closed algorithms built to drive conversions and ancillary revenue. Cannot be audited, cannot be used as official evidence.
Commercial data vendors	Cirium, OAG, ARC, IATA fare data, Apify and Bright Data scrapers	Selling competitive intelligence to airlines and travel companies	Expensive, licence-restricted, airline-facing rather than consumer-facing, and generally covers booked or filed fares

Category	Examples	What they are actually optimised for	Where they fall short for a price index
			rather than the full consumer out-of-pocket amount.
Aggregator internal data	MakeMyTrip, EaseMyTrip, Cleartrip	Maximising booking volume and commission	Genuinely rich data, but commercially confidential, single-channel, and structurally conflicted since the platform sets the very fees being measured.
Current official method	MoSPI manual price collection	Consistency with a decades-old survey framework	Monthly, counter-based, small sample, misses the online market and all dynamic pricing behaviour.

3.2 The scoring view

Scored against the criteria a national statistical office would apply. Yes means the capability exists in a usable form.

Capability	Meta-search	Predictors	Data vendors	Aggregators	MoSPI today	APIx
Daily frequency	Yes	Yes	Partial	Yes	No	Yes
Captures full out-of-pocket cost	No	No	Partial	Yes	No	Yes
Adjusts for product quality change	No	No	No	No	No	Yes
Offer-to-transaction correction	No	No	Partial	Yes	No	Yes
Published open methodology	No	No	No	No	Yes	Yes
Independently reproducible	No	No	No	No	Partial	Yes
Seasonally adjusted series	No	No	No	No	Partial	Yes
Maps to official CPI structure	No	No	No	No	Yes	Yes
Explains why the number moved	No	No	No	No	No	Yes
Free and public	Yes	Yes	No	No	Yes	Yes

4. Where the Real Gap Is

It is tempting to describe the gap as "nobody is scraping airfares for the government." That is not the gap. Scraping is the easy part and several teams will build it. The real gap sits in four places between a scraped number and a statistic a central bank can act on.

Gap 1 - Advertised price is not paid price

Every existing tracker records the price shown on a search page. But the price shown at a 1-day booking horizon is paid by only a small fraction of travellers; most people booked the same flight weeks earlier at a much lower fare. An index built on raw advertised quotes will systematically overstate airfare inflation.

Nobody bridges advertised price to realised price. This is the single largest gap, and it is the difference between a dashboard and an index.

Gap 2 - The product keeps changing shape

A price index assumes you are pricing the same thing over time. Airlines constantly change what a ticket includes: baggage allowance moves, fare families are renamed, seat selection becomes chargeable, departure timings shift. When the product changes, a naive index reads a price change that never happened.

Nobody separates genuine price change from product change. Statistical agencies consider this mandatory. No airfare tracker does it.

Gap 3 - The real inflation moved into the add-ons

Indian low-cost carriers have shifted a growing share of revenue from the base fare into ancillaries: checked bags, seat selection, meals, priority boarding, change and cancellation charges. A base fare can stay flat for a year while what a family actually spends to fly rises sharply.

Nobody measures the add-on layer as a price series. The inflation is real and currently invisible.

Gap 4 - Nothing is auditable

Official statistics carry a burden that commercial products do not. A published figure must be traceable, reproducible and defensible years later. Commercial platforms are closed by design and cannot meet this bar even if their numbers were perfect.

Nobody provides reproducibility. Without it, no scraped number can ever enter the CPI, no matter how accurate it is.

Our thesis

The bottleneck is not data collection. It is the missing statistical and governance layer that turns collected prices into an official statistic. APIx is built as that layer, with collection as the input rather than the product.

5. What We Will Build

APIx is a platform of six connected layers. Each one exists to close a specific gap identified above.

#	Layer	What it does
1	Collection layer	Gathers fare information every day across a defined basket of routes, carriers and booking horizons, following the full path to the point just before payment so that mandatory fees become visible. Operates under a transparent, declared, rate-limited and legally grounded access policy.
2	Decomposition layer	Breaks every collected quote into its real parts: base fare, fuel surcharge, airport and government levies, taxes, platform convenience fee, and each optional add-on. Nothing is left bundled.
3	Correction layer	The statistical core. Removes bad data, fills genuine gaps in a documented way, adjusts for changes in the product being sold, and converts advertised prices into estimated realised transaction prices.
4	Index layer	Combines corrected prices into published indices using internationally accepted index formulas, weighted by real passenger traffic and real booking behaviour. Produces daily, weekly and monthly series plus seasonally adjusted variants.
5	Intelligence layer	Adds the analysis that makes the number useful: why it moved, how much of the move is fuel cost, what is likely next week, and whether any route is behaving abnormally.
6	Delivery layer	Puts the output in front of each audience: an analyst dashboard for NSO and RBI, a machine-readable data service, a regulator alerting view, and a public transparency page.

5.1 What we publish

Output	Description
APIx (Headline)	The daily base-fare airfare price index. The direct successor to the current CPI airfare input.
TCT-APIx	Total Cost of Travel index. Includes all mandatory fees, taxes and a modelled typical add-on basket. This is what a traveller actually spends.
ANCx	Ancillary Price Index. Tracks baggage, seat, meal and change-fee pricing separately, so the add-on inflation story is visible on its own.
State-level series	The index broken out by origin state, so it can be fed directly into the state-wise CPI that NSO already publishes.
Seasonally adjusted series	The headline index with festival, holiday and wedding-season effects removed, so underlying trend is readable.
Fare Surge Alerts	Regulator-facing notifications when a route prices abnormally far above its own seasonal baseline.

6. Feature Scope

Features are grouped into three tiers. Tier 1 is what we commit to demonstrating. Tier 2 is built if time allows. Tier 3 is the documented roadmap that shows this is designed as infrastructure, not a one-off prototype.

6.1 Tier 1 - Committed for the prototype

Feature	What it delivers
Automated daily collection	Scheduled, unattended fare collection across the defined route basket, multiple carriers, multiple aggregators and five booking horizons, twice a day.
Full fare decomposition	Every quote separated into base fare, fuel surcharge, statutory airport fees, GST and platform convenience fee, with each add-on priced individually.
Data cleaning and outlier control	Removal of corrupt and non-reproducible quotes, with genuine demand spikes deliberately preserved rather than smoothed away.
Gap handling with disclosure	A documented method for handling sold-out flights and failed collections, with every filled value flagged so users can see exactly how complete the underlying data is.
Quality adjustment engine	Separates genuine price change from change in what the ticket includes. Our answer to Gap 2.
Offer-to-transaction correction	Converts advertised prices into estimated realised prices using booking-behaviour weighting. Our answer to Gap 1.
Index computation engine	Daily, weekly and monthly indices using internationally recognised formulas, with passenger-traffic and booking-horizon weighting.
Total Cost of Travel index	The parallel index that includes all mandatory and typical add-on costs. Our answer to Gap 3.
Movement attribution	For any day, a clear breakdown of which routes, carriers, fees and booking windows drove the index movement.
Validation against official data	A published comparison of our series against DGCA average fare data using a panel of accuracy measures, not a single number.
Analyst dashboard	Interactive time series, route heatmaps, booking-horizon curves and the attribution view, with role-based access.
Open data service	A documented, versioned machine-readable service delivering the index and its components for NSO, RBI and researchers.
Reproducibility ledger	Raw collected data preserved and tamper-evident, with the ability to regenerate any published figure exactly. Our answer to Gap 4.
Data quality monitoring	Continuous checks on coverage, freshness and source agreement, with automatic flagging of any day whose data does not meet the published standard.
Governance and legal framework	A documented access policy covering transparency of identity, rate limiting, off-peak operation, statutory basis, and a stated preference for official data-sharing arrangements over collection.

6.2 Tier 2 - Built if time allows

- **Ancillary Price Index (ANCx) as a standalone published series.** Add-on inflation tracked and reported independently.
- **Fuel passthrough decomposition.** Separates the portion of a fare rise caused by aviation fuel cost from the portion caused by demand.
- **Seasonal adjustment.** Removes predictable festival and holiday effects so the underlying trend is visible.
- **Short-horizon nowcast.** A one to two week ahead projection of the index with stated uncertainty, giving RBI a leading signal.
- **Fare surge alerting for DGCA.** Automatic flagging of abnormal route-level pricing during disruptions and festivals.
- **Public transparency page.** A simple citizen-facing view showing whether a fare on a route is high or low against its own recent history.

6.3 Tier 3 - Roadmap

- Expansion from the demonstration route basket to national coverage including regional and UDAN routes.
- Cross-modal substitution tracking against rail and long-distance bus fares on the same city pairs.
- International route coverage for the outbound travel component of the CPI.
- Formal data-sharing agreements with carriers, moving collection from observation to supplied feeds.
- Extension of the same platform to other fast-moving price categories such as hotels, ride-hailing and e-commerce, forming a general real-time price index platform for MoSPI.

6.4 Explicitly out of scope

- Booking, payment or any transaction on behalf of a user. APIx observes prices, it never purchases.
- Collection of any personal or passenger data. The system handles prices and flight attributes only.
- Consumer-facing price prediction intended to influence purchase timing. That is a commercial product category and a conflict of interest for a statistical system.
- Replacing the CPI itself. APIx supplies one improved input to an existing official process.

7. How We Are Different and Better

Our differentiation is not that we collect more data. It is that we do four things after collection that nobody else does, and one thing around collection that nobody else can.

Dimension	What everyone else does	What APIx does
Price definition	Records the advertised search price, because that is what drives clicks.	Records the complete amount a traveller pays, including every mandatory fee, and then corrects it toward the price actually transacted.
Product consistency	Compares this month's cheapest fare to last month's cheapest fare, whatever those tickets include.	Holds the product constant, so a reported price rise is a real price rise and not a quietly removed baggage allowance.

Dimension	What everyone else does	What APIx does
Purpose	Built to sell tickets, subscriptions or intelligence. The incentive is conversion.	Built to measure. There is no product to sell, so there is no reason for the number to lean in any direction.
Transparency	Closed proprietary algorithms. You are asked to trust the output.	Fully published methodology, open formulas, and raw data preserved so anyone can check the result themselves.
Accountability	Numbers change silently with no history and no explanation.	Every figure is versioned, every revision is recorded with a reason, and any movement can be attributed to its causes.
Access posture	Evasive scraping designed to look like an ordinary user and avoid detection.	A declared, identifiable, rate-limited public-interest collector with a statutory basis and an explicit preference for official data sharing.
Integration	Standalone dashboards that live outside the statistical system.	Output shaped to slot into the existing CPI structure, including state-level breakdown and standard statistical exchange formats.

7.1 The point most teams will miss

Many competing prototypes will treat anti-detection as the hard technical problem and will present proxy rotation and fingerprint masking as achievements. For a system carrying a government seal, that posture is a liability rather than a feature. A ministry that regulates the aviation sector cannot simultaneously run concealed infrastructure against it.

APIx inverts this. We collect openly, identify ourselves in every request, limit our rate well below any burden threshold, operate in off-peak hours, and position collection as a temporary bridge to formal data-sharing arrangements backed by existing statutory powers. This is both the defensible position and the one that actually scales, because a declared collector can negotiate access while a concealed one can only be blocked.

8. How We Stand Against Others

Positioned on the two axes that matter for this problem: how completely the true consumer price is captured, and how defensible the output is as an official statistic.

System	Price completeness	Statistical validity	Transparency	Fit for official use
Google Flights / Skyscanner	Low	None	Low	No
Hopper / Ixigo predictors	Low to medium	None	None	No
Commercial data vendors	Medium	Low	Low	Limited and costly
Aggregator internal data	High	Low	None	Conflicted
MoSPI manual survey	Low	High	High	Yes, but incomplete
APIx	High	High	High	Yes

Read down the last column. Only two options are usable by a statistical office at all, and only one of those actually measures the market where tickets are sold. That is the position APIx occupies, and it is empty today.

8.1 Why this position is hard to copy

- **Commercial players cannot move here.** Publishing an open, auditable methodology would expose the pricing intelligence they sell, and measuring total cost honestly would highlight the very fees they charge.
- **Pure engineering teams cannot reach here.** The barrier is statistical method and governance design, not scraping capability. Quality adjustment and offer-to-transaction correction are economics problems.
- **The value compounds with time.** A price index becomes more valuable the longer its history runs. A system that starts collecting properly today owns a dataset nobody can retroactively recreate.

9. Core Innovations and USPs

Six innovations define the product. The first three are our defensible differentiators; the remaining three make the system credible and usable.

USP 1 - Offer-to-Transaction Correction

The flagship innovation. Advertised fares are not what most people pay, because most people booked earlier. We model how bookings are actually distributed across the booking window and use that to convert the advertised price surface into an estimated realised price. Calibrated against official DGCA fare data.

Why it matters: This is the difference between a number that is interesting and a number that is correct. It is also the single most likely reason a scraped index would otherwise be rejected by NSO.

USP 2 - Quality-Adjusted Pricing

We treat an airline seat as a bundle of attributes rather than a single item: departure time, number of stops, baggage allowance, refundability, fare family, carrier. A statistical model separates the effect of these attributes from genuine price movement, so the index reports real inflation and not a change in what is being sold.

Why it matters: This is standard practice in national statistical offices worldwide and is entirely absent from every commercial airfare tracker. It is the most rigorous element of the product and the one an economist on a review panel will recognise immediately.

USP 3 - Total Cost of Travel measurement

We publish the base fare index and the full out-of-pocket index side by side. The gap between the two is a measurable finding: the portion of airfare inflation that current official statistics simply do not see because it lives in fees and add-ons rather than in the headline fare.

Why it matters: This is not just a feature. It is evidence. It quantifies the size of the blind spot the project exists to close, which is the strongest possible argument for the project itself.

USP 4 - Movement Attribution

Whenever the index moves, the platform explains the move: how much came from each route, each carrier, each booking horizon, each fee category, and how much is attributable to fuel cost as opposed to demand. The user sees a decomposition rather than a single unexplained number.

Why it matters: A central bank does not act on a number, it acts on a cause. Attribution converts a statistic into a policy input.

USP 5 - Reproducible by Design

Raw collected data is preserved in tamper-evident storage, the methodology is versioned, revisions follow a published policy, and any figure we have ever published can be regenerated exactly from stored inputs by an independent party.

Why it matters: Reproducibility is the entry requirement for official statistics. Without it, accuracy is irrelevant. With it, APlx can survive audit, parliamentary question and academic scrutiny.

USP 6 - Sovereign-grade Collection Governance

A collection policy built for a government institution: declared identity on every request, published access policy, conservative rate limits, off-peak operation, no personal data, a statutory basis for collection, and a clear migration path from observation to formal data-sharing agreements with carriers.

Why it matters: This is the difference between a prototype and something that can actually be deployed. It is also a direct answer to the first hard question any serious reviewer will ask.

The one-line pitch

APlx measures what Indians actually pay to fly, adjusts it the way a national statistical office must, explains every movement, and can prove every number. Nothing else in the market does more than the first half of the first clause.

10. Success Criteria

The prototype is considered successful if it meets the following, all of which are demonstrable within the build window.

#	Criterion	How it is demonstrated
1	A continuous daily index is produced with no manual intervention	At least 30 consecutive days of automated output
2	The index agrees with official DGCA fare data within a stated tolerance	A published validation panel covering correlation, level error, bias and ranking, not a single figure
3	The total-cost index is measurably above the base-fare index	A quantified and explained gap between the two published series
4	Any published figure can be recomputed by a third party	A live demonstration regenerating a past day's number from stored raw data
5	Any index movement can be explained	A live attribution breakdown for a selected day
6	Data quality is disclosed rather than hidden	Coverage, freshness and gap-fill rates published alongside every index value
7	The output fits the existing CPI process	State-level breakdown and standard statistical exchange format delivered through the data service

10.1 Principles we will not compromise on

- **Measure, never influence.** The system observes prices and never transacts, books or nudges behaviour.
- **Disclose uncertainty.** Where data is incomplete or estimated, we say so on the face of the output rather than in a footnote.
- **Open by default.** Methodology, formulas and aggregate data are public. Trust is the product.
- **Collect responsibly.** Identified, rate-limited and off-peak, with formal data sharing as the destination rather than an afterthought.